Nature Porter

I. Gameplay Introduction

```
    The purpose of this experiment is to stack four different colored building
blocks from bottom to top in the order of blue-green-red-yellow on the middle
gray block, then run the code. The robotic arm will execute in the following
order: grab the fourth layer block and place it in the yellow area, grab the
third layer block and place it in the red area, grab the second layer block and
place it in the green area, and grab the bottom layer block and place it in the
blue area. The arrangement of building blocks is shown in the figure below:
```

After executing the code, the robotic arm will move the building blocks to corresponding positions. The final effect is shown in the figure below:

II. Start Docker

After entering the main board desktop, open a terminal and run the following command to start the corresponding Dofbot container:

```
./Docker_Ros.sh
```

Then enter the following command in the Docker terminal to start jupyter-lab,

```
cd
jupyter-lab --allow-root
```

Then enter the main board's IP address followed by :9999 in the browser,

```
IP:9999         // Example: 192.168.1.11:9999
```

III. Code Content

Code path: /root/Dofbot/3.ctrl_Arm/10.move_block.ipynb

```
#!/usr/bin/env python3
#coding=utf-8
import time
from Arm_Lib import Arm_Device
# Create robotic arm object
Arm = Arm_Device()
time.sleep(.1)
```

```
# Define block clamping function, enable=1: clamp, =0: release
def arm_clamp_block(enable):
     if enable == 0:
         Arm.Arm_serial_servo_write(6, 60, 400)
     else:
         Arm.Arm_serial_servo_write(6, 130, 400)
        time.sleep(.5)
# Define robotic arm movement function, simultaneously control servos 1-5
movement, p=[S1,S2,S3,S4,S5]
def arm_move(p, s_time = 500):
     for i in range(5):
        id = i + 1
        if id == 5:
            time.sleep(.1)
            Arm.Arm_serial_servo_write(id, p[i], int(s_time*1.2))
         else :
            Arm.Arm_serial_servo_write(id, p[i], s_time)
        time.sleep(.01)
     time.sleep(s_time/1000)
# Robotic arm moves upward
def arm_move_up():
     Arm.Arm_serial_servo_write(2, 90, 1500)
     Arm.Arm_serial_servo_write(3, 90, 1500)
     Arm.Arm_serial_servo_write(4, 90, 1500)
     time.sleep(.1)
```

```
# Define variable parameters for different positions
p_mould = [90, 130, 0, 0, 90]
p_top = [90, 80, 50, 50, 270]
p_Yellow = [65, 22, 64, 56, 270]
p_Red = [117, 19, 66, 56, 270]
p_Green = [136, 66, 20, 29, 270]
p_Blue = [44, 66, 20, 28, 270]
p_layer_4 = [90, 72, 49, 13, 270]
p_layer_3 = [90, 66, 43, 20, 270]
p_layer_2 = [90, 63, 34, 30, 270]
p_layer_1 = [90, 53, 33, 36, 270]
```

```
# Move the robotic arm to a preparation position for grabbing
arm_clamp_block(0)
arm_move(p_mould, 1000)
time.sleep(1)
# Move the fourth layer building block to the yellow area
arm_move(p_top, 1000)
arm_move(p_layer_4, 1000)
arm_clamp_block(1)
arm_move(p_top, 1000)
arm_move(p_Yellow, 1000)
arm_clamp_block(0)
time.sleep(.1)
arm_move_up()
arm_move(p_mould, 1100)

# time.sleep(1)
```

```
# Move the third layer building block to the red area
arm_move(p_top, 1000)
arm_move(p_layer_3, 1000)
arm_clamp_block(1)
arm_move(p_top, 1000)
arm_move(p_Red, 1000)
arm_clamp_block(0)
time.sleep(.1)
arm_move_up()
arm_move(p_mould, 1100)

# time.sleep(1)
```

```
# Move the second layer building block to the green area
arm_move(p_top, 1000)
arm_move(p_layer_2, 1000)
arm_clamp_block(1)
arm_move(p_top, 1000)
arm_move(p_Green, 1000)
arm_clamp_block(0)
time.sleep(.1)
arm_move_up()
arm_move(p_mould, 1100)

# time.sleep(1)
```

```
# Move the first layer building block to the blue area
arm_move(p_top, 1000)
arm_move(p_layer_1, 1000)
arm_clamp_block(1)
arm_move(p_top, 1000)
arm_move(p_Blue, 1000)
arm_clamp_block(0)
time.sleep(.1)
arm_move_up()
arm_move(p_mould, 1100)

# time.sleep(1)
```

```
del Arm # Release the Arm object
```